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Software Requirements Specification
for
**RouteX: A Truck Profitability and Delivery Operations
Management System**

Version 1.0

Prepared for:
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Revision History

Name	Date	Reason For Changes	Version
RouteX	09/26/2026	Initial SRS Draft	1.0

1. Introduction

1.1. Document Conventions

This Software Requirements Specification (SRS) uses consistent headings and numbering to organize the requirements of the RouteX system. **Shall** indicates a required system behavior, **should** indicate a recommended behavior, and **may** indicate an optional behavior. **REQ** identifies functional requirements, **NFR** identifies

nonfunctional requirements, and **TBD (To Be Determined)** identifies information that has not yet been finalized. Requirement statements are written in clear and testable terms to describe what the RouteX system shall provide.

1.2. Purpose

The purpose of this Software Requirements Specification is to define the requirements and expected functions of **RouteX: A Truck Profitability and Delivery Operations Management System** for Redeemed Route Logistics Inc.

RouteX is a web-based system designed to help Redeemed Route Logistics Inc. manage trucking delivery operations and monitor truck profitability through centralized operational and financial records.

The system will provide authorized personnel with functions for recording and managing delivery transactions, monitoring delivery activities, recording revenue and expenses, calculating delivery profitability, searching and retrieving records, generating operational and financial reports, and managing secure system access.

The system is intended to improve the company's current manual processes, which primarily rely on Microsoft Excel, basic bookkeeping, and other manual methods for managing operational and financial information.

1.3. Project Scope

1.3.1. Scope

RouteX is a web-based system developed for Redeemed Route Logistics Inc. to improve the recording, monitoring, retrieval, and reporting of trucking delivery and financial activities. The project covers the following:

1. **Delivery Management** – Record and manage delivery details, including truck, driver, client, route, date, status, revenue, expenses, and POD, with automatic delivery margin calculation.
2. **Transactions Management** – Record and manage company expenses not directly related to a specific delivery.
3. **Search and Record Retrieval** – Search and retrieve delivery and transaction records using relevant information such as delivery ID, plate number, driver, client, and date.
4. **Reports and Profitability Analysis** – Generate delivery, revenue, expense, and profitability reports with printable and export functions.
5. **Security and Administration** – Provide login authentication, role-based access, user permissions, audit trails, and activity logs.

1.3.2. Limitations

1. RouteX is a partially automated system, and users are responsible for encoding and updating transaction and delivery information.

2. Delivery monitoring depends on information entered by authorized staff and does not provide automatic real-time GPS tracking.
3. The system does not automatically obtain location information from trucks or tracking devices.
4. The accuracy of records and reports depends on the accuracy and completeness of the information entered by users.
5. Features or integrations outside the agreed project requirements are not included in the current project scope.

1.4. Product Perspective

Redeemed Route Logistics Inc. currently relies on manual processes, including Microsoft Excel, basic bookkeeping, and group chats, to manage operational and financial information. The company does not currently have a formal integrated information system for its logistics operations.

RouteX is intended to provide a centralized web-based system for managing delivery operations and truck profitability. Authorized personnel can enter and manage delivery and expense information, which the system processes for delivery monitoring, profitability calculation, record retrieval, and report generation.

RouteX is **partially automated**. Users remain responsible for entering and updating information, while the system assists in organizing, processing, calculating, retrieving, and reporting the recorded data.

The proposed system uses:

- **Frontend:** Next.js, React, TypeScript, Tailwind CSS
- **Backend:** Next.js Server Actions + API Routes
- **Database:** PostgreSQL through Supabase
- **ORM:** Prisma
- **Authentication:** NextAuth.js/Auth.js
- **Deployment:** Vercel

1.5. Product Functions

RouteX will provide the following main functions:

1. **Delivery Management** – Manage delivery records, delivery status, revenue, delivery expenses, and POD documents.
2. **Transactions Management** – Record and manage company expenses not associated with a specific delivery.
3. **Search and Record Retrieval** – Search and retrieve delivery, transaction, and related records.
4. **Reports and Profitability Analysis** – Generate operational, financial, and profitability reports.
5. **Security and Administration** – Manage authentication, user access, permissions, and activity logs.

1.6. Design and Implementation Constraints

The following constraints apply to the development and operation of RouteX:

- The system shall be implemented as a **web-based application** accessible through a modern web browser.
- The frontend shall use **Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, and Lucide React**.
- The backend shall use **Next.js Server Actions + API Routes**.
- The database shall use **PostgreSQL through Supabase**.
- The system shall store uploaded files and supporting documents using **Supabase Storage**.
- Authentication shall use **NextAuth.js (Auth.js)**.
- The system shall be deployed using **Vercel**.
- The system requires a **stable internet connection**.
- The minimum hardware requirement is a **dual-core 2.0 GHz processor or higher and 8 GB RAM**.
- Access shall be restricted to **authorized users**.
- Delivery information shall depend on **staff-entered updates** and shall not use automatic GPS or tracking-device integration.

1.7. User Documentation

The following user documentation will be provided with RouteX:

1. **User Manual** – A step-by-step guide covering login, delivery management, transaction management, search and record retrieval, report generation, and other functions based on the user's assigned role.
2. **System Guide** – A brief guide describing the main pages, features, and navigation of RouteX.
3. **Training Materials** – Supporting materials for user training and system turnover.

The documentation will be provided in **digital format** and may include screenshots and step-by-step instructions. It will be based on the final implemented version of RouteX.

1.8. Assumptions and Dependencies

The system is based on the following assumptions and dependencies:

- Authorized users will provide accurate and complete delivery and expense information.
- Users will keep their login credentials confidential.
- Users will have access to a supported web browser and stable internet connection.
- The client will provide the necessary information and resources for development and implementation.

- The client will participate in system testing and User Acceptance Testing (UAT).
- RouteX depends on the availability of its required hosting, database, authentication, and internet services.
- Changes or additional features outside the agreed scope require approval from both parties.

2. Data Requirements

2.1. Data Validation Rules

- **Delivery ID, truck plate number, driver name, and client name** shall be required when creating a delivery record.
- **Delivery dates** shall use a valid date format.
- **Revenue and expense amounts** shall be numeric and non-negative.
- **Delivery margin** shall be automatically calculated from revenue and delivery expenses and shall not be directly editable.
- **Delivery status** shall be limited to predefined values, such as Pending, In Transit, Delivered, and Cancelled.
- **Proof of Delivery (POD)** uploads shall use supported file types and size limits and shall be linked to an existing delivery record.
- **User accounts** shall use unique usernames or email addresses, and passwords shall follow the required security rules.

2.2. Data Security and Privacy Requirements

- The system shall implement **authentication and role-based access control** so that only authorized users can access and manage records.
- Passwords shall be stored using a **one-way cryptographic hash** and shall not be stored or displayed in plain text.
- Data transmitted between the client browser and server shall be **encrypted in transit using HTTPS/TLS**.
- The system shall comply with the **Data Privacy Act of 2012 (RA 10173)** and applicable data protection requirements.
- User activities, including record creation, modification, deletion, and login events, shall be recorded through **audit trails and activity logs**.
- Access to client data shall be limited to purposes related to the **development, maintenance, and operation** of RouteX and shall be kept confidential.

2.3. Data Retention and Backup Requirements

- Delivery, transaction/expense, user, and POD records shall be retained to support operational use, historical record retrieval, and reporting.
- RouteX data shall be backed up to reduce the risk of data loss due to system failure or other incidents.
- The specific **data retention period and backup schedule are TBD** and shall be finalized based on the client's requirements and system implementation considerations.

- Upon termination of the project or agreement, client data shall be handled according to the agreed terms between the client and provider.

3. System Features

3.1. Functional Requirements

3.1.1. System Feature #1

3.1.2. System Feature #2

3.2. Nonfunctional Requirements

3.2.1. Performance Requirements

3.2.2. Safety Requirements

3.2.3. Security Requirements

3.2.4. Software Quality Attributes

3.2.5. Business Rules

4. System Model and Diagrams

4.1. Use Case Diagrams

4.2. Entity Relationship Diagrams

4.3. Activity Diagrams

5. Other Requirements

Appendix A: Glossary

Appendix B: To Be Determined List

Appendix C: References

Requirements Validation / Consistency Review